

#20 Customer Insights Engine

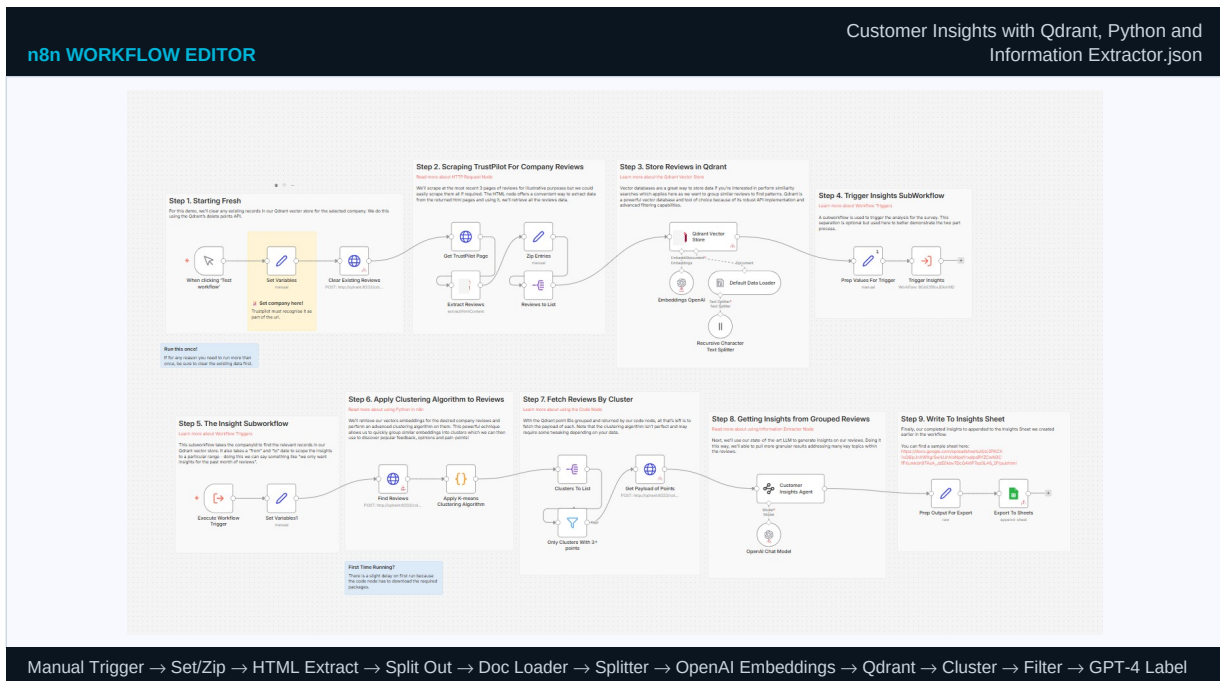
ML / Customer Intelligence

Unstructured reviews to structured business intelligence automatically every week

BUSINESS PROBLEM

Companies collect thousands of customer reviews and support tickets but can never read them all. This workflow scrapes reviews, embeds and clusters them in Qdrant, and produces a weekly insight report with themes, sentiment scores, and representative quotes — from thousands of comments to a one-page report in 15 minutes.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Manual trigger — weekly review ingestion run

OUTPUT — Weekly insight report row per theme — written to Sheets

```
{
  "company_id": "company-001",
  "trustpilot_url": "https://trustpilot.com/review/example.com",
  "date_from": "2024-05-15",
  "date_to": "2024-06-15",
  "max_pages": 50,
  "cluster_count": 20
}
```

```
{
  "json": {
    "week_ending": "2024-06-15",
    "reviews_processed": 3847,
    "clusters_found": 18,
    "themes": [
      {"theme": "Delivery Speed", "mentions": 342, "sentiment": 0.31, "top_quote": "Ordered 5-day delivery, arrived day 9. Third time this month."},
      {"theme": "Product Build Quality", "mentions": 287, "sentiment": 0.91, "top_quote": "Solid build, feels genuinely premium — worth the price."},
      {"theme": "App Checkout UX", "mentions": 198, "sentiment": 0.28, "top_quote": "Checkout crashes on Android after entering card details."}
    ],
    "processing_time_sec": 847
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Manual or Weekly Cron trigger	Sources: Trustpilot scrape, Zendesk ticket export, App Store reviews. Configured per company in workflow settings.
2	■ Set/Zip — normalise across sources	Standardises fields from all sources: {text, source, rating_1_5, date, customer_segment} for consistent processing.
3	■ HTML Extract — scrape review pages	html node extracts review text and rating from Trustpilot paginated listing. Handles up to 50 pages automatically.
4	↑ Split Out — individual reviews	Each review split into a separate n8n item for parallel processing through the embedding pipeline.
5	■ Document Loader — text cleaning	Strips HTML tags, emojis, and normalises unicode. Preserves star rating as metadata on each chunk.
6	⌘ Recursive Splitter — chunk long reviews	Reviews above 500 chars split at sentence boundaries. Short reviews kept whole. Rating metadata preserved.
7	■ OpenAI Embeddings — vectorise	text-embedding-3-small: 1536-dim vectors. Batched at 100 reviews per API call for cost efficiency.
8	■ Set Variables — Qdrant payload	Constructs upsert payload: {id, vector, payload:{text, source, rating, date, customer_segment}}.
9	■ HTTP — Qdrant k-means clustering	k=20 clusters requested. Returns cluster_id assignment per review point. Similar complaints naturally group.
10	■ Filter — clusters with 3+ reviews	Removes singleton noise clusters. Keeps only statistically meaningful theme clusters for GPT-4 labelling.
11	↑ Split Out — per meaningful cluster	Each cluster processed separately so GPT-4 can generate a focused theme label and summary.
12	■ GPT-4 — label each cluster	Given cluster's top 10 reviews → generates: {theme_name, sentiment_score, summary, representative_quote, trend}.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI API Key	platform.openai.com	OpenAI API
Qdrant Cloud URL + API Key	cloud.qdrant.io → Create cluster	Qdrant API

SAMPLE RUN — what this workflow processed and produced

Scenario	A D2C consumer electronics brand receives 300-500 reviews per week across Trustpilot, Amazon, and their own website. The customer experience team had no systematic way to identify recurring issues.
Input received	Weekly cron triggered Sunday 06:00. Trustpilot scraper fetched 50 pages (3,847 reviews). Date range: last 30 days. Zendesk export added 412 support ticket summaries.
What the workflow found	4,259 total texts ingested, embedded, and clustered into 18 meaningful groups in 14 minutes 7 seconds. Top pain point: 'Delivery Speed' — 342 mentions with sentiment 0.31. Three reviews in this cluster mentioned the same carrier failing on promised dates.
Output produced	18-row insight report in Google Sheets. CX team actioned two items in the same week: switched secondary carrier after identifying pattern, and filed app crash bug report for Android checkout issue. Both resolved within 10 days.

JSON: Customer Insights with Qdrant, Python and Information Extractor.json	GitHub: trm-customer-insights-engine	qdrant.tech/
--	--	------------------------------